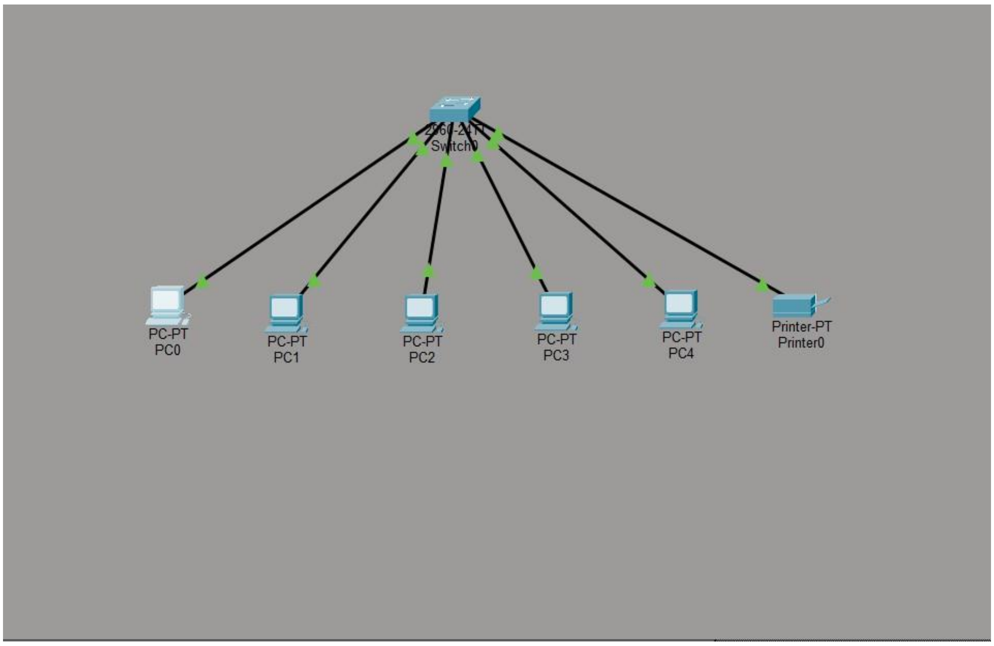
TASK 3:-

Build a LAN topology in packet tracer for 5 systems and 1 printer using a class c subnet.

To design and configure a Local Area Network (LAN) in Cisco Packet Tracer by connecting five computers and one printer using a Class C IP addressing scheme, and to verify network connectivity.

Network Topology

The network consists of one switch connected to five computers and one printer. All devices are connected using Copper Straight-Through cables.



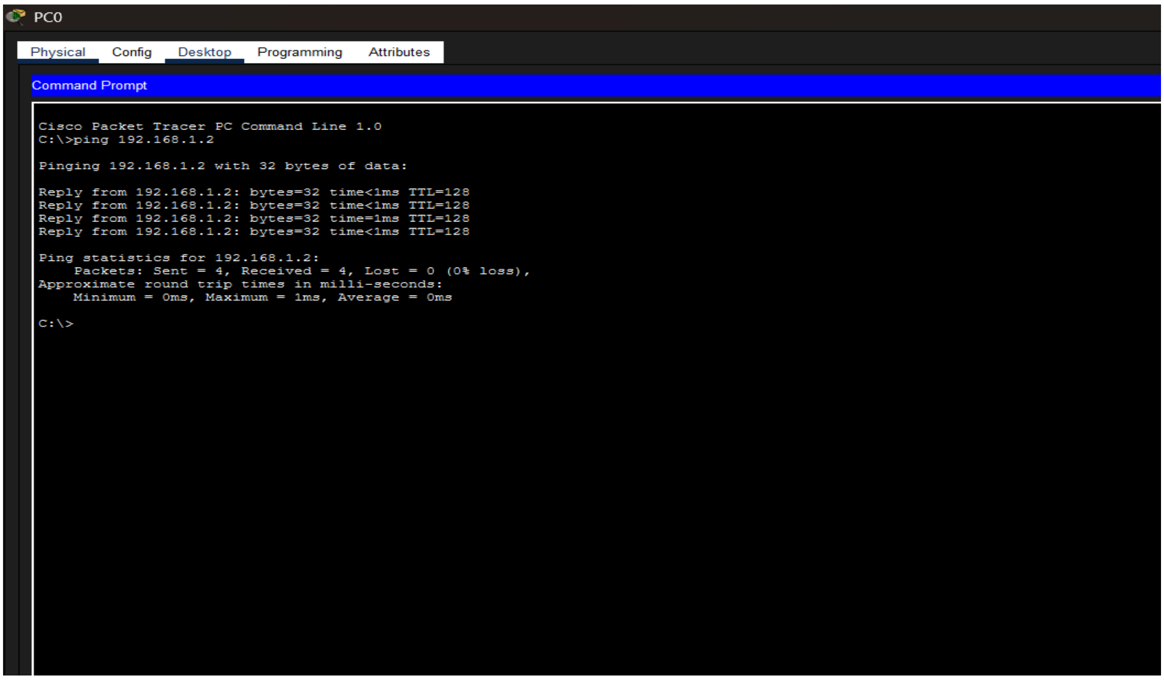
Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0
PC4	192.168.1.5	255.255.255.0
Printer	192.168.1.10	255.255.255.0

Network Address: 192.168.1.0/24

Default Gateway: Not Required (No Router Used)

Procedure

1. Open Cisco Packet Tracer and create a new project.
2. Place one Cisco 2960 Switch in the workspace.
3. Add five PCs and one Printer from the End Devices section.
4. Connect all devices to the switch using Copper Straight-Through cables.
5. Configure the IP address and subnet mask for each device according to the IP addressing table.
6. Save the configuration.
7. Open the Command Prompt on PC0 and test connectivity using the **ping** command with the IP addresses of the other PCs and the printer.
8. Verify that all devices respond successfully, confirming proper LAN communication.



Observation

- All devices were assigned unique IP addresses within the Class C network (192.168.1.0/24).
- The switch successfully connected all devices.
- Ping tests between the computers and the printer were successful.
- The LAN operated correctly without requiring a router because all devices were on the same subnet.

The LAN topology consisting of five computers and one printer was successfully created in Cisco Packet Tracer using a Class C subnet (192.168.1.0/24). All devices communicated successfully through the switch, and the network connectivity was verified using the **ping** command.